Data Analytics

4-7-25
Friday.

* It is used to processing the raw data to draw the meaningful communications/insights. & improve business performance, reduce product cost, save time, to make better dec's
* It is divided into 4 types:-
1. Descriptive D.A 2. Diagnostic D.A
3. predictive D.A 4. prescriptive D.A

Descriptive :→ tells what happend in the past ?
Diagnostic :→ tells us why did it happened.? (to find reasons to failure)
predective :→ predict the future outcomes.
what will happen next ?
presciptive :→ What should we do.?

statistics (mean, median, mode)
Descriptive :→ gives the summary of the past data.
Diag :→ used to find the reasons to the failure.
⇓
hypothesis testing.
Predictive → ML Techniques (SVM, random forest techniques)
Presciptive ⇒ DL, NLP.

Data.

Structural data
↓
→ data has predefined structure
→ Represented as matrix (rows, cols)
→ we can search easily.
Ex:→ student details, dbms, mysql, spreadsheets, Excel.
→ we can do mathematical Oper^ns (statistics)

Unstructural data.
→ It doesn't have any predefined format.
→ search is difficult
Ex:→ social media.
→ we can't do calcul^ns. (statistics)

Data (Acc to statistics)

Qualitative data
→ It is represented in form of description
→ It is non numerical data.
→ c/s categorical data.
→ used to give labels to data

Quantitative data
→ It is in the form of numerical
→ we can measure the data
→ It is numerical & measurable.

Ex: good, bad, male, female, below avg, above avg

→ we can't measure the data

Ex: colour, gender, feedback etc.

→ we can't do statistics

→ collecting through surveys/ interviews

Ex: age, height, weight, speed, distance, no of students, etc.

→ we can do statistics

→ collecting through measuring scales.

Defn of Data Analytics:-

* Qualitative & quantitative techniques & processes are used to business Analytics to get meaningful conclusions, enhance productivity & profit gain.

(OR)

* Data Analytics is the process of examining raw data sets to draw conclusions, discover patterns, profit gain, actionable insights.

* Again in Quantitative data, we have 2 types.

Quantitative Data.

- discrete
  - → represent the whole number/ integer
  - Ex:- age.
- continuous.
  - → represent in the form of floating pt.
  - Ex:- ht, wt

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* **Frame work for data-driven decision making.**

→ It is used to improve business performance

1) Define problem/ make decision (we need to identify the problem & take decn)
2) collecting data (we need to collect data relevant to the taken decision)
   → Gathering relevant info from diff sources.
3) Analyze the data. (need to apply the statistics mean, mode, median).

(4) Interpret the results. (we need to transmit raw data into actionable insights)

(5) Implement the actions/decisions. (need to implement the framed decⁿ).

(6) Monitoring & Iteration. (check the business performance continuously).

* Data types in Measuring scale / Measuring scale types in statistics

* It is divided into 4 type. → Used to measure the data

(1) Nominal scale } qualitative
(2) ordinal scale }
(3) Interval scale } quantitative
(4) Ratio scale. }

* Nominal :⇒ Names of the variables (labeling).
* It is like categorical data. (labels).
Ex:- gender, colors etc.
* we can't do calculⁿs, only we can compare the data. (=, ≠)

(2) Ordinal scale :→ It follows the order.
Ex:- small, medium, large...
→ Ratings [1 2 3 4 5]
* It can compare the data (=, ≠, >, <).
* It can some period of time (Range).

(3) Interval scale :→ some period of time some interval.
Ex:- Temp calcⁿ in some interval.
* It can do comparisions and also addⁿs & subtⁿs (=, ≠, >, <, +, -).

(4) Ratio scale : fractions.
* The data is measurable.
* we can measure hts & wts.
* The data is continuous.
* It do compⁿs & all arithmetic operⁿs (=, ≠, >, <, +, -, *, /, %).
* It doesn't have -ve values.

	oper's (mode) Nominal	(median, mode) Ordinal	(mean, mode, median) Interval	(All calc's, statistics) ratio (Avg mark)
naming / labeling.	✓	✓ (good, bad)	✓ (1st, 2nd schedule)	✓
order	✗ (order for names)	✓	✓	✓
difference	✗	✗	✓ (last credit)	✓
absolute zero (meaningless) (there is no obs)	✗	✗	✗ (0 is meaningless)	✓ (there is no obs if wt=0, ht=0) (there is meaning for 0)

mode — most freq occurrent val
median — middle value
mean — avg value.

Nominal — Mode.
Ordinal — Mode, median.
Interval — Mean, median, mode.
Ratio — statistics & All Arithmetic oper's

} oper's.

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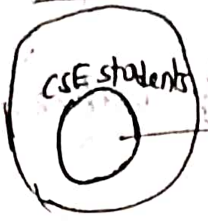
* Population & Sample

popul^n :→ whole / entire data
→ It includes all the observ's from data set
sample :→ It is specific grp from population / subset of popul^n.

* This sampling makes inferences about popul^n
* If we chose wrong sample, we get wrong result
* If we take right sample, we get right results.
* chosing sample is very important.
* surveying on sample is easier than on popul^n.
  we can take less time, cost, data etc...
* For popul^n, it is difficult, more cost, more money etc... & practically impossible.
* we can't apply statistics on entire popul^n;
  we can do statistics on sample & get results

* We can apply sample result to population.



popuⁿ (RGUKT students)
sample.

*Descriptive Data-Analytics

* It tells about what happened in the past / gives status/ summary about the past data.
* It will represent complex data into user-friendly manner. (converts)
* Here we can apply statistics & after it gives summary of past data.
* Techniques are 3 types :→
  1. Data Cleaning
  2. summarizⁿ of data by using descriptive statistics
  3. Data visualization.

(1) Data cleaning.

* we need to clean the data.
* 1. we need to find missing values in the data set. (we can fill with avg if any one is missing (h+1wt))
  we have to fill them with avg.
  2. If there are any duplicates, we need to remove from dataset.
  3. If there is any inconsistency in data, we need to correct the data.
* We are preprocessing the data before analysis of data. & next we will apply desciptive statistics which gives summary of data.

(2) Summarizⁿ of data

* It summarize data based on 3 things
  1. Distribution. 2. Central Tendency 3. Dispersion.

(1) Distribution:→ ~~Find the~~ measure freq of each individual value) in the data set.
observⁿs

2) Central Tendency: Used to measure central part of the data set.

3) Dispersion: calculate the range of values from the mean.

Central Tendency. (3 types)

1) Mean 2) Median 3) Mode

(Avg value → Used to measure avg value of the dataset of data set)

[sum of obs / total no of obs] for popul^n → (μ) → $\boxed{\mu = \frac{\sum_{i=1}^{N} x_i}{N}}$

for sample → ($\bar{x}$) $\boxed{\bar{x} = \frac{\sum x_i}{n}}$

Ex: 19, 20, 21, 22, 23, 24.

$\boxed{\text{mean } (\bar{x}) = \frac{\text{sum of observations}}{\text{no of observations}}} = \boxed{\frac{x_1 + \dots + x_n}{n}}$

$= \boxed{\frac{\sum_{i=1}^{n} x_i}{n}}$

* If freq is given → mean $(\bar{x}) = \boxed{\sum f_i \times \text{Value}}$

$= \boxed{\frac{\sum_{i=1}^{n} f_i * x_i}{n}}$ //

* Summ^n of all observ^ns deviations from mean is 0

i.e. $\boxed{\sum (x_i - \bar{x}) = 0}$ //

* If we want to find mean, we need to go through all elements of data set.

② Median
→ Middle value of the data set.
→ First we need to arrange data set in sorted order & then pick the mid value.
* This median will divide data set into 2 equal parts.

1 2 3 → $\frac{3+1}{2} = 2$

* If n is odd → $\boxed{\frac{n+1}{2}}$ //

If n is even → $\boxed{\left[\frac{\frac{n}{2} + \frac{n+2}{2}}{2}\right]}$ //

1 2 3 4 → $\frac{2+3}{2}$

$= 2.5$ //

* Here, no need to go to all values in data set
* Median is more stable than mean.

(3) Mode.

* Most frequently occuring value in the data set.

Ex:- 19, 20, 19, 19, 21, 22 → mode = 19.

* If there is no repeated value in data set, then there is no mode in data set.

* This all will gives summary of data in single value
* sometimes we need to calculate all values in data set.
* 1. Percentile. 2. decile 3. Quartile.

(1) Percentile:- Gives out of 100.
→ They will represent/ measure the posn of observn in the dataset.
→ It will divides data set into 100 equal parts
→ It is indicated with "P_x"
→ It is used to find posn of obs 'x'.
→ If we want to

∴ $\boxed{P_x = \frac{x(n+1)}{100}}$ (n = no of observ's
x = observn posn).

(2) Decile:- It is used to divide data set into 10 equal parts.

→ $P_{10} = \frac{10(n+1)}{100}$, $P_{20} = \frac{20(n+1)}{100}$ ... $P_{90} = \frac{90(n+1)}{100}$.

→ It will find posn of multiples of 10's posn.

(3) Quartile:- It will divide data set into 4 equal parts.

$Q_1 = P_{25} = \frac{25(n+1)}{100}$ $Q_3 = P_{75} = \frac{75(n+1)}{100}$

$Q_2 = P_{50} = \frac{50(n+1)}{100}$ $Q_4 = P_{100} = \frac{100(n+1)}{100}$.

* Central tendency is used to measure central part, Sometimes we need to go will all values, so we use percentile ...

③ **Dispersion:** used to measure spreading the values from the mean
→ It is of 4 types
① Range ② Inter Quartile range / Inter Quartile distance (IQR / IQD) ③ Variance
④ Std deviation.

① **Range:** Used to measure diff b/w max value & min value in dataset

$$\boxed{Range = max - min.}$$

Ex: 10, 20, 30, 40, 50, 60, 70, 80, 90
range = $90 - 10 = \boxed{80}$

② **Inter Quartile Range:** Used to measure diff b/w Quartile 3 & Quartile 1

$$\boxed{IQR = Q_3 - Q_1.}$$

Lower bounded = $Q_1 - 1.5 \times IQR$
Upper bounded = $Q_3 + 1.5 \times IQR$

$$Q_3 = \frac{75(9+1)}{100} = \frac{75(10)}{100} = \frac{75}{10} = 7.5 \text{ (posn)}$$

$$Q_1 = \frac{25(10)}{100} = 2.5 \text{ (posn)}$$

(take values)
IQR = $Q_3 - Q_1 = 70 - 20 = \boxed{50}$

③ **Variance:** The sumn of each obsern from mean is c/s variance.

population variance (σ^2) = $\boxed{\dfrac{\sum_{i=1}^{n}(x_i - \mu)^2}{n}}$

whole values.

sample variance (s^2) =

$$\boxed{s = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}}$$

sub part of populn.

④ **std deviation:** sqrt of variance

$$\boxed{sd = \sqrt{variance}.}$$

popul^n std deviation (σ) = $\boxed{\sqrt{\frac{\sum_{i=1}^{n}(x_i-\mu)^2}{n}}}$

Sample std deviation (s) = $\boxed{\sqrt{\frac{\sum_{i=1}^{n}(x_i-\bar{x})^2}{n-1}}}$

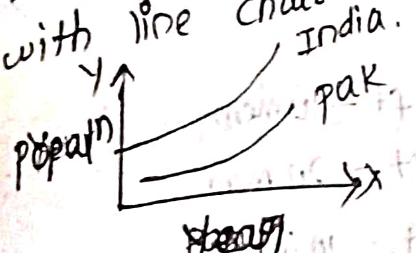
* These are all included in summariz^n of data.

3) Data Visualization.

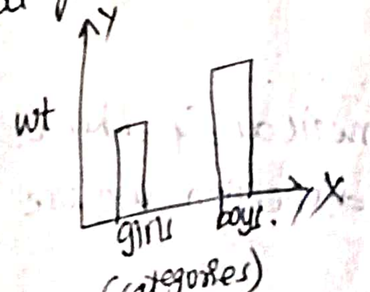
* Here the data is represented in:-
1. plots 2. charts 3. Maps.
* By this, everyone can understand the data easily.
* It is used for clear understanding.
* After applying statistics, the data which is in numerical format, it is represented in graphs, charts etc---

Charts:-
1. Line charts
2. bar charts
3. pie charts
4. scatter plot
5. Histogram.
6. box plot.

1. Line chart:- For time line series, we go with this.
→ when data is categorical & numerical we go with line chart.

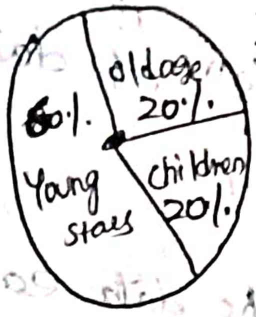


2. Bar chart:- Used when data is categorical & numerical & mainly used for comparision of diff categories.

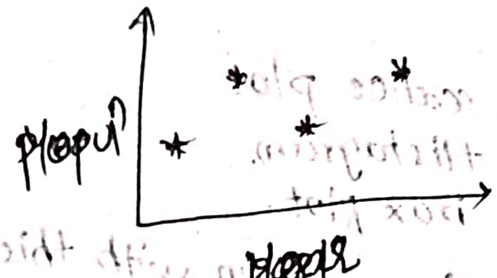


3. __pie chart__:→ represent proportion of the whole data
→ It is represented with circle.
→ Used when data is categorical, numerical & in proportion.

Internet Users

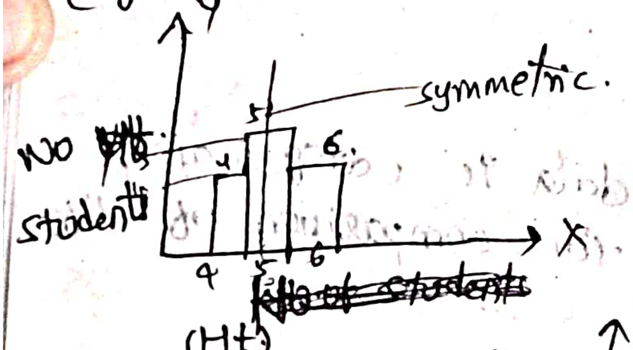


4. __Scatter plot__:→ similar to line chart & points are indicated with stars
* Used to measure corelation b/w variables.



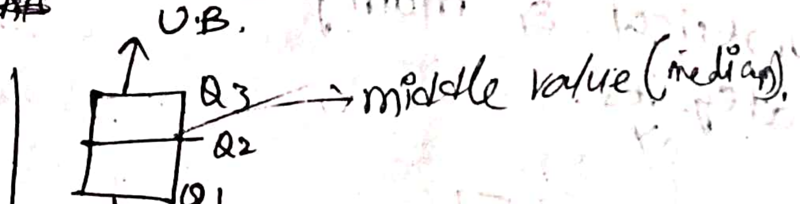
* correln value are in b/w 0 & 1.
near 0 → weak correln
near 1 → strong correln

(5) __Histogram__:→ when data is categorical, numerical, continuous & symmetric we use histogram.
(sym → left & right are equal portions from mean)



4 ft – 10 mem
5 ft – 20 mem
6 ft – 10 mem

(6) __BOX plot__:→



when data is categorical, numerical & when we want to measure distance / dispersion we use Box plot

Probability distribution

* **Random experiment** :- we can't predict the outcomes before running the program.

Ex:- Tosing coin, Rolling die.

* **Random variable** :- Which maps the real no for each sample in the sample space.

Ex:- Tossing a coin $S = \{HH, HT, TH, TT\}$.

No. of heads $= \{2, 1, 1, 0\}$ → random variables.

⇒ It is of 2 types
- 1. Discrete R.V ⇒ Takes only finite no's
- 2. Continuous R.V. ⇒ Takes infinite no's (floating pts)

D.R.V ⇒ Rolling die, Tossing coins.
C.R.V ⇒ Time taken to fill Water tank.

* **Probability Distribution function**

⇓

* possible value of sample is c/s "prob. distn".

* It is of 3 types
  - → probability Mass fn (PMF)
  - → prob. Density fn (PDF)
  - → Cumulative Distn fn. (CDF)

* **prob. Mass function**

⇒ Here we take discrete Rand variables.

→ It satisfies 2 condns :→ $\boxed{\begin{array}{l} 1)\ f(x) \geq 0. \\ 2)\ \sum f(x) = 1. \end{array}}$

Ex:- $S = \{HH, TH, HT, TT\}$.
$S = \{2, 1, 1, 0\}$.

$f(x) = P(X = x_i)$

$P(X=0) = P(T, T) = 1/4 \geq 0$

$P(X=1) = 2/4 = 1/2 \geq 0$

$P(X=2) = 1/4 \geq 0$.

$\sum f(x) = \frac{1}{4} + \frac{2}{4} + \frac{1}{4} = \frac{4}{4} = \boxed{1}$.

∴ It is PMF//.

* **prob. Density function.**

→ Here we take continuous rand variables.

→ It satisfies 2 condⁿs

$\int_a^b f(x)\,dx = 1 \rightarrow a \le x \le b.$

$$\boxed{\begin{array}{l} 1)\ f(x) \ge 0. \\ 2)\ \int_{-\infty}^{\infty} f(x)\,dx = 1. \end{array}}$$

③ Cumulative Dist. fⁿ.

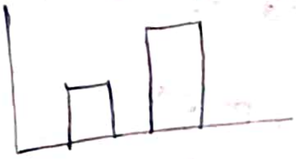
→ we use both discrete & continuous variables.

① Discrete probability distⁿ	2. Continuous prob distⁿ
1. Binomial → 2 possible o/p are possible T/F. → n times bernoulli.	1. Normal.
2. Bernoulli	2. Exponential.
3. poisson. → $n \to \infty$, $p \to 0$. ($n > 100$)	3. Student T distⁿ.
4. Uniform → same prob for each outcome.	4. chi-square.

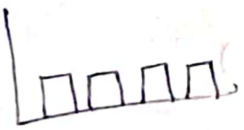
① Binomial :→

④ poisson.

② Bernoulli :→.



③ Uniform :→

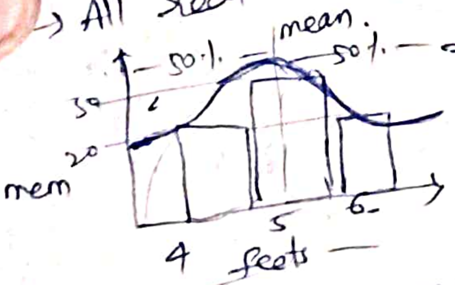


* Normal Distribution.

→ when data is symmetric & continuous: we use this distⁿ.

→ All real world problems, follows Nrml distⁿ.

→ Distⁿ of data is centered around the mean.



→ Here, mean = median = mode at center pt.

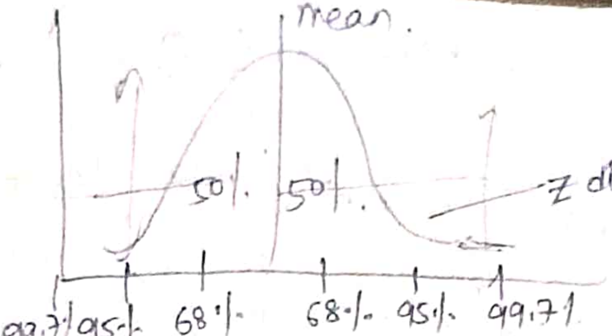
→ It allows some statistics to Analyze the data

→ The shape of the graph is "bell shaped"

→ This shape is depends on the "variance"

→ Total area under the curve = 1 = 100%.

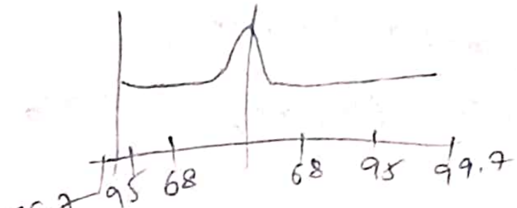
→ shape of distⁿ depends on "Variance". < sharp (sma... / flat (sma...)



3 parts :→ 68%
95%
99.7%

* The max data will be covered at mean → at "68%" (center part)

* In T dist, max data will be covered at Tail part



⇒ Max data will be covered at Tail part.

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{\frac{-(x-\mu)^2}{2\sigma^2}} \Rightarrow \boxed{\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}}$$

$\boxed{z = \frac{x-\mu}{\sigma}}$

μ = mean

σ = st deviation = $\sqrt{var}$ //

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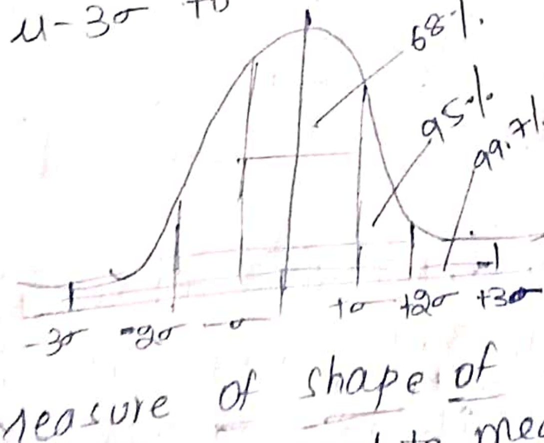
* 3 empheerical rules :⇒

1. $\mu-\sigma$ to $\mu+\sigma$ ⇒ first std deviation. (most data covered here)
2. $\mu-2\sigma$ to $\mu+2\sigma$ ⇒ 2nd std deviation.
3. $\mu-3\sigma$ to $\mu+3\sigma$ ⇒ 3rd std deviation



→ 1st std devⁿ covers ⇒ 68% data

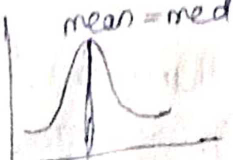
→ 2nd std devⁿ covers → upto 95% data

→ 3rd std devⁿ covers → upto 99.7% data.

* Measure of shape of distribution

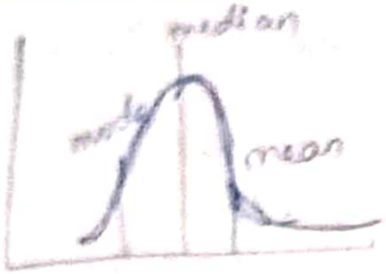
1) skewness → used to measure symmetricness of data
2) kurtosis → used to measure tailedness (peak/flat) of data.

skewness :→



⇒ There is no skewness (skewness = 0)

→ It follows exactly Normal distⁿ.

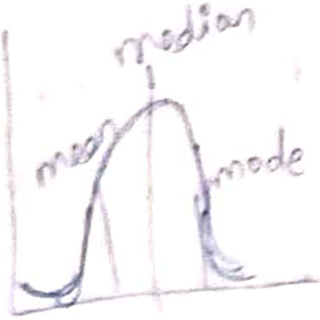


→ $\boxed{\text{skewness} > 3\left(\frac{\sigma}{\sqrt{n}}\right)}$ std error

→ It is +ve / right skewed

$\boxed{\text{mode} < \text{median} < \text{mean}}$

→ Right side tail part is more



→ $\boxed{\text{skewness} < 3\left(\frac{\sigma}{\sqrt{n}}\right)}$

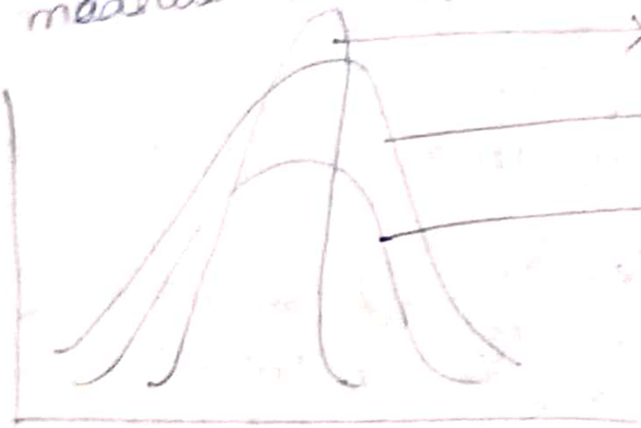
→ It is -ve / left skewed.

$\boxed{\text{mean} < \text{median} < \text{mode}}$

→ left side tail part is more

* Kurtosis

→ To measure Tailed part of the distribution.



→ Leptokurtic ($\beta > 3$)

→ Normal distn (mesokurtic) ($\beta = 3$)

→ platokurtic. ($\beta < 3$)

* It is denoted by 'β'.

$\beta = 3$ (mesokurtic)

$\beta > 3$ (Leptokurtic)

$\beta < 3\left(\frac{\sigma}{\sqrt{n}}\right)$ (platokurtic).

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① CGPA of students 3.36, 1.56, 1.48, 1.43, 2.64, 1.48,
2.77, 2.80, 1.38, 2.84, 1.88, 1.83, 1.87, 1.95, 3.43,
2.57, 3.74, 1.98,
1.28, 3.67, 2.83, 1.71, 1.68, 1.66, 1.66, 2.96, 1.77,
1.62, 2.74, 3.35, 1.80, 2.86, 3.28, 1.14, 1.98, 2.96,
3.75, 1.89, 2.16, 2.07.

① calculate the mean, median, mode & std deviation

② calculate the interquartile range. (IQR)

3) Create a Histogram for the data.

2. Unit

Sampling:- subset of the population

Sampling:- It is a process of selecting subset of observ's from the popul^n to make the inferences abt the parameters (mean, st dev^n, proportion)

* For popul^n ⇒ it is time taking, more cost & can't apply statistics for entire popul^n

→ for sample → takes less time, less money & easy to apply statistics.

* Sampling Estim^n:- It will helps to dec^n making in surveys. Estimate inferences abt popul^n.

1. point estimation

It gives the single value of popl^n/sample

Ex:- ($\bar{X} = 80$) (sample mean)

∴ $\mu \sim 80$. (pop^n mean)

→ sometimes it fails.

2. Interval Estimation.

→ It gives range of values of pop^n/sample.

Ex:- $\mu \pm \sigma$, $\mu \pm 2\sigma$. → gives intervals

Ex:- [75 to 85] ($\bar{X}$ range)

[75 to 85] [for μ range)

→ Interval est^n is better than pt estim^n.

If interval is wide ⇒ more confidence b|t less precision

→ If interval is narrow → less confidence, more precision.

* Sampling Process.

1) define Target Population

→ First we need to identify particular category on which we want to do survey. (students, public etc..)

2) determine sampling frame.

→ It is mainly used for identifying source of data, where data is available.

3) Choose sampling method.

→ It is very important.

→ It is classified in 2 categories
- probabilistic sampling method
- Non-probabilistic sampling method

(1) Probabilistic sampling method

→ It follows the prob and the distn.

(1) Random sampling method.
(2) stratified " " → cost effective
(3) cluster " " → gives accurate results
(4) systematic " "
(5) Bootstrap " "

(1) Random sampling method

→ When data is homogeneous, we will use random samp. method. (every obsern has equal priority)
→ It is very popular & most frequently used method.

(2) Stratified sampling method   popln → sample

→ When popuLn data is non-homogeneous, we go with this.
→ It will divide total populn into some grps.
→ Each grp is c/s "stratum"   populn → stratum
→ each stratum should be "homogeneous".
→ We have to select samples randomly after dividing into stratums.
* Based on stratum size, we need to select sample

$\boxed{\text{sampling seln} \propto \text{stratum size}}$

→ From all stratums, we need to collect the sample
→ After selecting the samples, we need to combine all samples to get the final sample.

Ex:- $s_1 \times n + s_2 \times n + s_3 \times n + s_4 \times n = N$. (n = sample each stratum)

* After that we can do statistics to make inference

(3) Cluster sampling method

* It is similar to stratified " (data is non-homogeneous, cluster - homogeneous)
* It divides total popln into some clusters.
* Each grp is c/s "cluster" & from each cluster

we need to select the samples

cluster

* Here, No need to include all clusters in final sample but in stratified, we need to include all stratified samples to final sample.

i.e $\underbrace{C_1 \times n + C_2 \times n + C_3 \times n}_{\text{(majority clusters)}} = N$. [we ignored C_4]

* After getting final sample, we can do statistics to make inferences [They convert heterogenous sample → homogeneous]

④ ~~Bootstrap~~ systematic sampling Method

→ we can select the samples → k^{th} element sample from the populn

→ In each grp, we need to select k^{th} element sample.

⑤ Bootstrap sampling Method

* It is mainly used in ML algorithms.
* The samples are replacing in each ~~tp~~ time.
* Each time we need to take sample & apply ~~computing~~ algorithm.
* These follows prob & distribution

② Non-Probabilistic Sampling method.

* It doesn't follow any prob & distribn.

① Convience sampling method
② voluntary " "

→ less cost
→ ~~may be the~~
→ can't say about accurate results.

① convience sampling method

* In convience, here we select the nearby samples to do survey.
* Here we choose nearest approach & avoiding farthest approaches.

② voluntary sampling method

* Here, choosing the sample voluntarily. (Independent samples)
* choosing sampling method is very important.

④ Decide sample size.

→ we need to chose sample size

→ If we take large sample size ⇒ the accuracy may be high.

→ If we take less sample size ⇒ less accuracy.

* The accuracy will be depends on sample size.

⑤ Select the samples

* we need to select the samples from total populn. from sampling frame (used to identify source of data).

* from sampling frame, we need to select the samples.

⑥ collect the data

* After selecting the samples, we need to collect the data related to the sample.

⑦ Analyze & make the inferences about population.

* After selecting & collecting sample, we need to apply statistics on sample.

* This sampling is used to make inferences abt. popuLn.

---

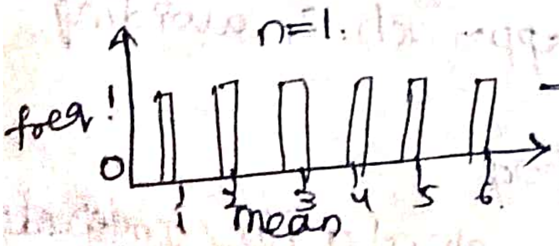
* Sampling Distribution.

* popuLn may or may not follow Normal distn.

* sampling distn must follow the Normal distn.

Ex:- If we roll a die $n=1$. (1 time)

$p = 1/6$ for $\{1, 2, 3 \dots 6\}$



→ uniform distribution

$$\text{mean} = \frac{1+2+3+4+5+6}{6} = \frac{\text{Sum of obs}}{\text{no of obs}}$$

$$\boxed{\bar{x} = \frac{21}{6} = 3.5}$$

$\bar{x} = 3.5$.

$\sigma = 1.7$.

$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{1.7}{\sqrt{1}} = 1.7$

$$\sigma^2 = \frac{\sum (x_i - \bar{x})^2}{n} = \frac{(1-3.5)^2}{6} + \dots$$

$$\boxed{\sigma = 1.7.}$$

for n=2 (when 2 dies are rolled at once)

total Per^n's = 36.

Mean

$(1,1) = \frac{1+1}{2} = 1$ (2,1) = 1.5 (3,1) = 2 (4,1) = 2.5

(1,2) = 1.5 (2,2) = 2 (3,2) = 2.5 (4,2) = 3

(1,3) = 2 (2,3) = 2.5 (3,3) = 3 (4,3) = 3.5

(1,4) = 2.5 (2,4) = 3 (3,4) = 3.5 (4,4) = 4

(1,5) = 3 (2,5) = 3.5 (3,5) = 4 (4,5) = 4.5

(1,6) = 3.5 (2,6) = 4 (3,6) = 4.5 (4,6) = 5

(5,1) = 3 (6,1) = 3.5

(5,2) = 3.5 (6,2) = 4

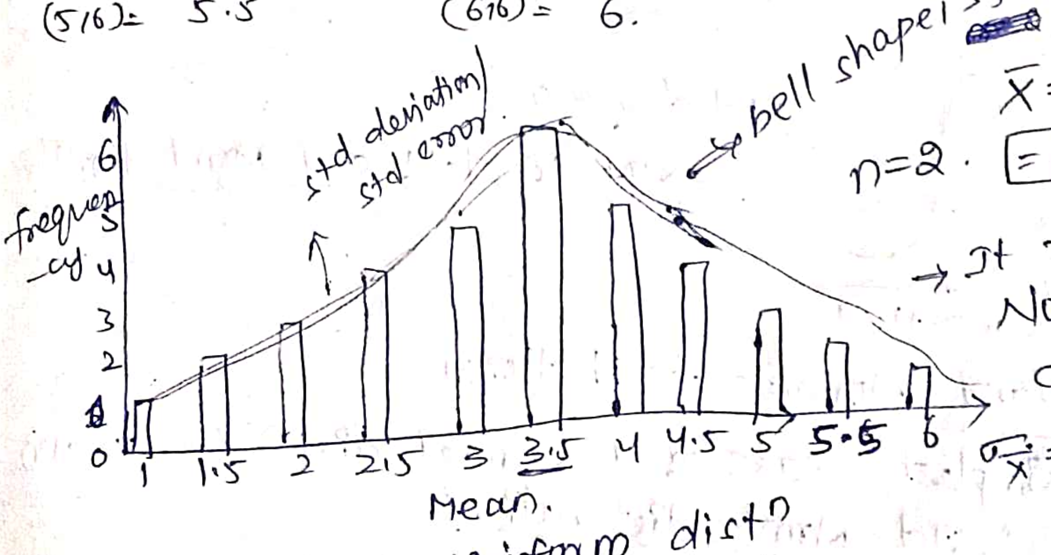
(5,3) = 4 (6,3) = 4.5

(5,4) = 4.5 (6,4) = 5

(5,5) = 5 (6,5) = 5.5

(5,6) = 5.5 (6,6) = 6.

(mean for every probability).



$\bar{X} = \frac{1+1.5+ -- +6}{11}$

n=2. $\boxed{= 3.5}$

→ It follows Normal distribution

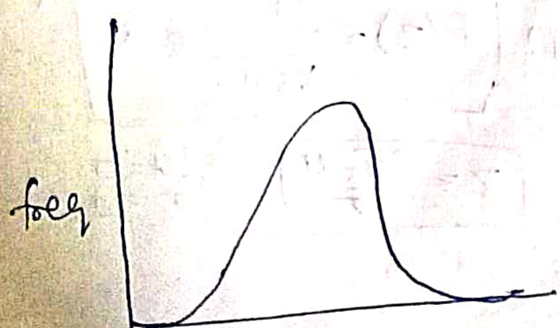
$\sigma_{\bar{x}} = \frac{1.7}{\sqrt{2}} = \boxed{1.2}$

* When n=1 ⇒ Uniform dist^n

When n=2 ⇒ Normal dist^n

when n=3 (216) ⇒ Normal dist^n (peak ↑ses)

$$\boxed{\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \text{std error}}$$



for n = 3. $\bar{x} = 3.5$.

$\sigma_{\bar{x}} = \frac{1.7}{\sqrt{3}} = \boxed{0.9}$

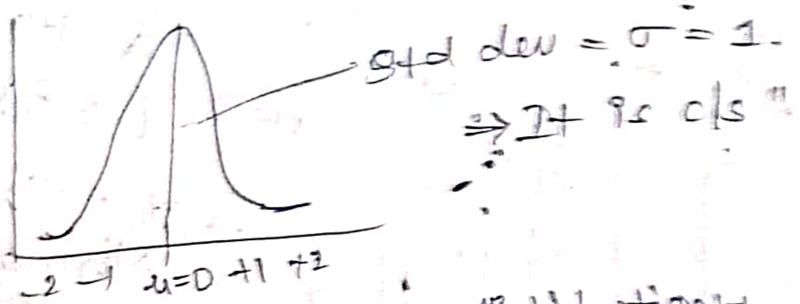
$n=4$.

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{1.7}{\sqrt{4}} = \boxed{0.6}$$

* When we take large samples, then std error is decreases

* $\boxed{n \propto \frac{1}{\sigma_{\bar{x}}}}$ $\boxed{\begin{array}{l} n\uparrow, \ \sigma_{\bar{x}}\downarrow \\ n\downarrow, \ \sigma_{\bar{x}}\uparrow \end{array}}$

* That's why we have to take large samples.
* when we take large samples, it follows Normal distn & error is also decreased.
* The mean ≈ populn mean
* When $n\uparrow$, there will be a correct peak, ~~then~~ When $\mu=0$, std devn $=1$, it is c/s "std Nrml distn"



⇒ It is c/s "std Nrml distn"

* Standard Normal distribution:-
→ The simplest case of a normal dist° is known as the "std Nrml distribution".
* This is a special case when $\mu=0$ (mean) & σ (std devn) $= 1$. & it is described by "probability density function" $f(x)$

$$\boxed{f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}} \quad (\text{or}) \quad \boxed{f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}}$$

$$\boxed{z = \frac{\bar{x}-\mu}{\sigma}} \quad \begin{array}{l}\mu=0\\ \sigma=1\end{array}$$

* Central limit Theorem (CLT)

* It is essential thing in statistics

* It is mainly used in Hypothesis Testing

* It tells that sampling $dist^n$ mean ~~or~~ $popul^n$ $dist^n$ mean

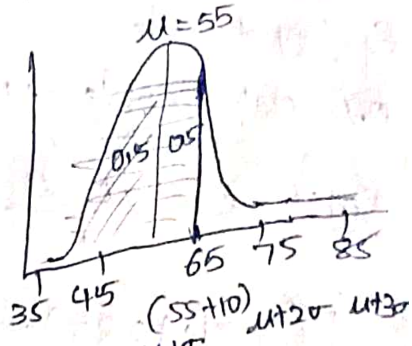
* so when we take large sample ($n \geq 30$), we can apply "central limit Theorem".

* when sample size ($n < 30$) is small, it doesn't follow normal $dist^n$, so we can't apply CLT.

* It describes abt sampling dist mean is approx equal to the $popul^n$ $dist^n$ mean. $\boxed{\bar{x} \simeq \mu}$

Ex:- Normally distributed test scores with mean $\mu = 55$ & $\sigma = 10$. calculate the probability of randomly picked test score below 65.

Sol:- $Z = \frac{\bar{x} - \mu}{\sigma} = \frac{65-55}{10} = \frac{10}{10} = \boxed{1.}$



$\therefore f(z) = \frac{1}{\sqrt{2\pi}} \cdot e^{-\frac{1}{2}}$

$P(\bar{x} < 65) = 0.5 + P(z=1)$

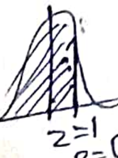
$= 0.5 + 0.34$

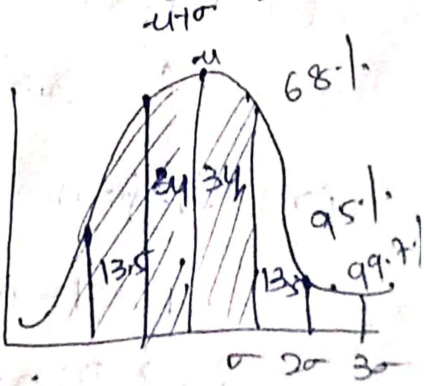
$= \boxed{0.84}$ //

$P(z < 1) = \cancel{0.5 + P(0.274)}$

$= 0.5 + 0.34$

$= \boxed{0.84}$ //.





$\frac{95-68}{2}$

Ex:- suppose it knows the ht of all canadians follows as norml $dist^n$ with a mean of 160 cm & std dev^n of 7 cm what is prob that avg ht of 10 random canadians is less than 157 cm.

$Z = \dfrac{\bar{x} - \mu}{\left(\frac{\sigma}{\sqrt{n}}\right)} = \dfrac{157 - 160}{\left(\frac{7}{\sqrt{10}}\right)}$ = (If n is given)

$Z = \dfrac{\bar{x} - \mu}{\left(\frac{\sigma}{\sqrt{n}}\right)} = \dfrac{157 - 160}{\left(\frac{7}{\sqrt{10}}\right)} = \dfrac{157 - 160}{2.22} = -1.35$

$P(X < 157) = P(Z < -1.35) =$

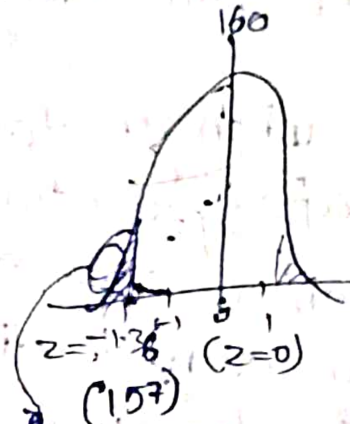
$0.5 - P(\cancel{\ldots})$

$0.5 - \cancel{0.41149}$

$0.5 - 0.08891$

$\cancel{= 0.41149.}$

$\boxed{P(Z < -1.36) = 0.08691}$ //.



p=0.086.

(z table contains values -∞ to that value).

(3) suppose it knows that the hts of all canadians follows a normal distⁿ with a mean of 160 cm & std deviation of 7 cm, what is the proportion of all people that have hts > 170 cm.

Solⁿ $z = \dfrac{\bar{x} - \mu}{\sigma} = \dfrac{170 - 160}{7} = \dfrac{10}{7} = 1.4285 = 1.43$

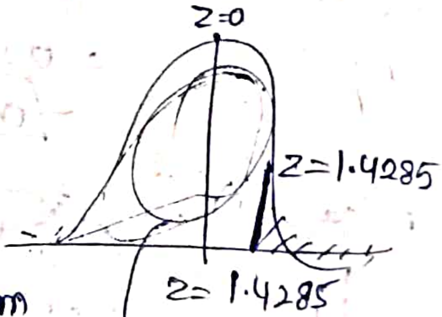
$P(Z > 1.4285) =$

$\cancel{0.5}\left(\cancel{0 < Z < 1.43}\right)$

$P(Z = 1.43) = \cancel{0.5}\ 0.92364$ (ie from $-\infty$ to 1.43 = 0.92364

$P = 1 - 0.92364$

$\boxed{P = 0.07636}$ //.



p=0.92364

b/t we want from 1.43 to ∞ ∴ 1 − 0.92364)

* Estimation of population parameters

* $\mu, \sigma, \hat{P}$, shapes of distⁿ → populⁿ parameters
* we can't directly estimate populⁿ parameters, we have to estimate through sampling values
* we estimate this in 2 ways:-

1. point estimation — gives single value for entire popⁿ → may not be correct
2. Interval estimation
   ↳ gives interval, where populⁿ parameter value lies in that interval → gives exact accuracy

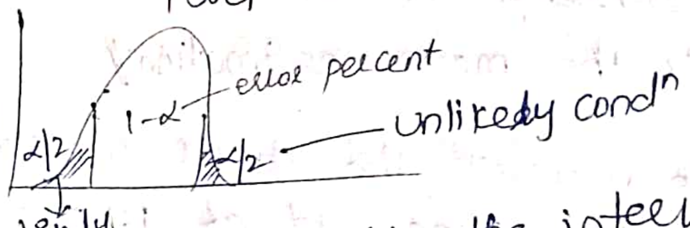
* These range of values are c/s "confidence intervals"

$\boxed{\mu \in [\bar{x} - E, \bar{x} + E]}$ E = margin of error

* Margin of error (E) = $\boxed{Z_{\alpha/2}\left(\frac{\sigma}{\sqrt{n}}\right)}$ → std error

$Z_{\alpha/2}$ → confidence level

If $Z_{\alpha/2} = 95\%$
i.e 95% of values are true.



* The populⁿ mean should be within the intervals.

Confidence Interval

$$C.I = \bar{x} \pm Z_{\alpha/2}\left(\frac{\sigma}{\sqrt{n}}\right)$$

$$C.I = \left[\bar{x} - Z_{\alpha/2}\left(\frac{\sigma}{\sqrt{n}}\right) < \mu < \bar{x} + Z_{\alpha/2}\left(\frac{\sigma}{\sqrt{n}}\right)\right]$$

lower bound, upper bound

* populⁿ mean should be b/w l.b & u.b.

Confidence level	Z-score ($Z_{\alpha/2}$)	** α
90%	1.645	0.10
95%	1.96	0.05
98%	2.33	0.02
99.9%	2.576	0.01

① A sample of 100 students has an avg ht of 160cm. The populn std devn is 10cm. Find 95% confidence interval for the avg ht.

Sol:- $\boxed{C.I = \bar{x} \pm Z_{\alpha/2}\left(\frac{\sigma}{\sqrt{n}}\right)}$ $Z_{95\%} = 1.96$

$= 160 \pm 1.96\left(\frac{10}{\sqrt{100}}\right)$

$= 160 \pm 1.96\left(\frac{10}{10}\right)$

$\boxed{C.I = 160 \pm 1.96}$

$\boxed{C.I = [158.04, 161.96]}$ → we are taking interval for populn mean from sampling mean

Conclusion:-

We are 95% confident that the true avg ht of all students is b/w 158.04 cm & 161.96 cm.

This is the mean estimation.

② We measure the hts of 40 randomly chosen men and get a mean ht of 175 cm. we also known the std. devn of men's ht is 20 cm. calculate the 95% confidence intervals.

Sol:- $C.I = \bar{x} \pm Z_{\alpha/2}\left(\frac{\sigma}{\sqrt{n}}\right)$ $Z_{95\%} = 1.96$

$= 175 \pm 1.96\left(\frac{20}{\sqrt{40}}\right)$

$= 175 \pm 1.96 \times 3.163$

$= 175 \pm 6.2$

$\boxed{C.I = [168.8, 181.2]}$

Conclusion:- we are 95% confident that the true avg ht of all men's is b/w 168.8cm & 181.2cm

(3) There are 100's of apples on the trees. so you randomly chose just 46 apples & get a mean of 86 & std devn is 6.2. so calculate 95% confidence Interval of apples size.

Sol:- $C.I = \bar{x} \pm Z_{\alpha/2}\left(\frac{\sigma}{\sqrt{n}}\right)$ $Z_{\alpha/2} = 1.96$

$= 86 \pm 1.96\left(\frac{6.2}{\sqrt{46}}\right)$

$= 86 \pm 0.91 \times 1.96$

$\boxed{C.I = \cancel{[85.09, 86.91]}}$ //

$C.I = 86 \pm 1.784$

$\boxed{C.I = [84.216, 87.784]} \approx \boxed{C.I = [84.22, 87.78]}$ //

Conclusion:- Avg size of apples is lies b/w 84.216 & 87.78 //

(4) Amount of time spent by 20 students on an online course is given below, Assuming that the proportion of time spent follows a Nrml distn & std deviation = 3.1 hrs. Calculate the 90% of confidence Interval for the mean time spent by the students

→ sample time spent by students on online course :-
4.7, 9.3, 8, 7.4, 9.2, 1.7, 7.2, 8.6, 9, 6.9, 9.2, 11.2, 7.6, 4.9, 5.3, 2.8, 12.3, 10.6, 5.7, 3.8.

Sol:- $\bar{x} = \frac{4.7 + 9.3 + 8 + \cdots + 3.8}{20}$ $Z_{90\%} = 1.645$

$\bar{x} = 7.27$

$C.I = \bar{x} \pm Z_{\alpha/2}\left(\frac{\sigma}{\sqrt{n}}\right) = 7.27 \pm 1.645\left(\frac{3.1}{\sqrt{20}}\right)$

$= 7.27 \pm 1.645 \times 0.69$

$= 7.27 \pm 1.135$

$$\boxed{C.I = [6.135, 8.405]}$$

Conclusion:- The mean time spent by students is b/w <u>6.135 hr & 8.405 hr</u>